

Practical extraction and reporting language (Perl)

Nathan Warner



Northern Illinois
University

Computer Science
Northern Illinois University
United States

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Perl

- **Practical extraction and reporting language (perl):** Perl is a high-level, general-purpose, interpreted, dynamic programming language. Though Perl is not officially an acronym, **Practical Extraction and Reporting Language** is commonly used as a **backronym**.
- **use version pragma:** The use VERSION pragma in Perl enforces a minimum version required to run a script and automatically enables modern language features

You should place this declaration at the very top of your file. Modern Perl code uses the vX.Y v-string format rather than old-style decimals:

```
0 use v5.36
```

Versions v5.10 and higher automatically activate the feature pragma for that release. For example, use v5.36; is equivalent to loading use feature ':5.36';. This gives you immediate access to built-in keywords like say, state, and updated signatures without typing extra code

Declaring use v5.11 or higher automatically turns on the strict pragma (use strict;) lexically across your file.

Declaring use v5.35 or higher automatically turns on the warnings pragma (use warnings;

- **use feature:** In Perl, the use feature pragma allows you to enable newer syntactic features and capabilities without breaking backward compatibility for older programs. Because it is a lexical pragma, any feature you enable is only active from the line it is declared until the end of its enclosing code block.
 - **Individual Features:** Pass a list of feature names to the pragma.

```
0 use feature 'say';  
1 say "Hello, world!"; # Appends a newline
```

- **Feature bundles:** Load all features made available in a specific Perl release by prefixing the version with a colon

```
0 use feature ':5.36'
```

- **Implicit loading:** Requiring a minimum Perl version automatically enables the feature bundle for that version
- **Command Line Execution:** Use the -E flag instead of -e to quickly test code snippets with all features enabled for your current Perl version.

```
0 perl -E 'say "Fast one-liner testing!'"
```

- **Using the diagnostics pragma:** Another pragma that you may find useful is the diagnostics pragma. This translates the normally terse diagnostics emitted from the perl compiler and the perl interpreter into much more useful ones

```
0 use diagnostics;
```

The diagnostics pragma makes your warnings much more verbose, and it slows the start-up time of your script considerably. You should remove it before putting your code into production.

- **Memory management:** Perl 5 memory management is primarily based on reference counting, with a runtime that manages Perl values such as scalars, arrays, hashes, and objects for you.
- **Compiled?:** Every time you execute a script, the Perl interpreter first compiles your source code into an internal bytecode before immediately running it. Because this compilation happens seamlessly in the background right before execution, Perl is often colloquially categorized as an interpreted scripting language.
 - **Compilation Phase:** The engine parses your source code. It converts text into optimized opcode structures.
 - **Execution Phase:** The internal Perl virtual machine runs that bytecode.

Comments

- **Standard comments:** Uses a hashtag

```
0  # Comment
```

- **Block comments:** We can also use Plain Old Documentation (POD)

```
0  =begin comment
1  ...
2  ...
3  =end comment
4
5  =cut
```

=cut resumes Perl code execution. "Stop ignoring this text; switch back to executable Perl code."

=end comment closes a =begin comment region. "This specific comment block is done, but continue treating subsequent text as documentation."

- **__END__**: You can signal to Perl the end of your program by using the special **__END__** tag. Anything below **__END__** will be ignored by Perl.

```
0  #!/usr/bin/perl
1
2  print "Hello world";
3
4  __END__
```

Command line options

- **Warnings:** Turn on warnings with -w

```
o  #!/usr/bin/perl -w
```

Variables

- **Pragmatic information (pragma):** A code pragma (short for "pragmatic information") is a special instruction or command embedded in source code. It tells a compiler, interpreter, or assembler how to process the program, but it does not change the actual logic or output of the software itself.
- **Note about variables:** In Perl, it is not necessary to declare your variables before you begin. You can summon a variable into existence simply by using it, and it will be globally available to any routine in your program. If you're used to programming in C or any of a number of other languages, this may seem odd and even dangerous to you. This is indeed the case. That's why you want to use the strict pragma.
- **Variable sigils (prefixes):** In Perl, variable prefixes are called sigils, and they immediately identify the data type of the variable.
 - \$ used for scalars
 - @ used for arrays
 - % used for hashes. A hash is an unordered collection of key-value pairs (associative arrays).
- **my, our, and state:** In Perl, my declares a truly private lexical variable, state creates a persistent static variable that initializes only once, and our creates a temporary shortcut to a global package variable

A lexical variable is a variable that is available only within a specific, textually defined part of a program, such as a function, a block, or a file.

- **The strict pragma:** Once we use the strict pragma, we have to explicitly declare new variables with scope keywords my, our, or state.

```
0  #!/usr/bin/perl -w
1  use strict;
2
3  my $x = 50; # Okay
4  $y = 100; # Bad
```

- **Variables with name a or b:** \$a and \$b are special variables in Perl. they are implicitly used by Perl's built-in sort function. Inside the sort block, Perl automatically places the two values being compared into \$a and \$b. Consequently, strict 'vars' specifically permits \$a and \$b without requiring my
- **Scalars:** A scalar is a single item of data

```
0  my $name = "Arthur";
1  my $whoami = 'Just Another Perl Hacker';
2  my $meaning_of_life = 42;
3  my $number_less_than_1 = 0.000001;
4  my $very_large_number = 3.27e17; # 3.27 by 10 to the power of
   ↪ 17
```


A scalar can be text of any length, and numbers of any precision (machine dependent, of course). Perl doesn't need us to tell it what type of data we're going to put into the scalar. In fact, Perl doesn't care if the type of data in the scalar changes throughout your program. Perl magically converts between them when it needs to

- **Single vs double quotes:** In Perl, quotes are required to distinguish strings from the language's reserved words or other expressions. Either type of quote can be used, but there is one important difference: double quotes can include other variable names inside them, and those variables will then be interpolated

```
0 my $x = 50;
1 print "$x";
```

- **Special characters:** Special characters, such as the `\n` newline character, are only available within double quotes. Single quotes will fail to expand these special characters just as they fail to expand variable names
- **String concatenation:** Uses the dot notation

```
0 print "Hello, " . "World!";
```

- **Bash style interpolation:**

```
0 my $x = "Hello";
1 print "${x}, World";
```

- **References / dereferencing:**

```
0 my $x = 10;
1 my $y = \$x; # Creates a reference
2
3 $$y = 25; # $$ Dereferences
```

- `\$scalar`
- `\@array`
- `%hash`
- `&function`

- **Typeglobs:** In Perl, a typeglob (represented by the asterisk `*` sigil) is an internal data type that holds a complete symbol table entry for a given identifier.

Because Perl allows you to use the exact same name for different variable types simultaneously (e.g., `$apple`, `@apple`, `%apple`, and `sub apple`), Perl uses a typeglob to bundle all of these variables into a single slot in the package's symbol table. You can visualize a typeglob as a structure containing separate internal pointers for each variable type

Slot / Prefix	Data Type	Example
\$	Scalar	\$apple
@	Array	@apple
%	Hash	%apple
&	Subroutine / Code	apple()
*	IO / Filehandle	APPLE

- **Typglob uses:** Assigning one typglob to another maps all variables of that name to the new name.

```

0  *foo = *bar;
1  # Now $foo is an alias for $bar, @foo for @bar, %foo for
   ↪ %bar, etc.

```

By assigning a reference to a typglob slot, you can alias just one specific type

```

0  *foo = \$bar;
1  # Only $foo and $bar are now the same variable.
2  # @foo and @bar remain completely separate.

```

- **Creating constants**

```

0  my $PI = \3.14;

```

creates a reference to a read-only scalar constant containing the value 3.14.

```

0  print $$PI;
1  $$PI = 10; # Error

```

- **Creating constants with typglob:** You can generate read-only variables by assigning a reference to a constant value inside a typglob.

```

0  *PI = \3.14159;
1  $PI = 4; # Compiles to a run-time error: "Modification of a
   ↪ read-only value..."

```

- **Legacy filehandles:** In older Perl code, typglobs or references to typglobs (*FH) were required to pass filehandles into subroutines. Modern Perl handles this automatically via lexical scalar filehandles (open my \$fh, ...).
- **Slots in a typglob:** Use

```
0  *NAME{SLOT}
```

Common slot names include:

```
0  SCALAR
1  ARRAY
2  HASH
3  CODE
4  IO
5  FORMAT
6  GLOB
```

- **Inspecting prototypes:** If you need to retrieve or inspect a prototype programmatically from an existing function, you can use the built-in prototype function:

```
0  my $proto = prototype('my_subroutine_name');
```

Arrays

- **Creating arrays:** Arrays are initialised by creating a comma separated list of values:

```
0  my @arr = (1,2,3);
```

Perl's arrays, like those in many other languages, are indexed from zero

- **Accessing:** As you would expect

```
0  print $fruits[0];
```

We use the dollar sign here because the indexing produces a scalar.

- **Slicing:** If we wanted to only deal with a portion of the array, we use what we call an array slice. These are written as follows:

```
0  @arr # entire array
1  @arr[1,3]
2  @arr[0..2]
```

- **Negative indexing:**

```
0  print $arr[-1] # 3
```

- **Array size:**

```
0  print $#arr + 1;
```

Note that `$#arr` gets the index of the last element. We can also use

```
0  my $size = @arr;
1  print my $size = @arr;
```

- **qw// (quote words):** It takes whitespace separated words and turns them into a list

```
0  my @arr = qw/hello , world/
```

Your list can stretch over multiple lines, and your delimiter doesn't need to be a slash. Whatever punctuation character that you place after the `qw` becomes the delimiter. So if you prefer parentheses over slashes, that's no problem at all

- **Printing out arrays**

```

0 my @arr = (1,2,3);
1 print @arr;
2 print join(", ", @arr);
3 print "@arr"

```

The first method takes advantage of the fact that `print` takes a list of arguments to print and prints them out sequentially. The second uses `join()` which joins an array or list together by separating each element with the first argument. The third option uses double quote interpolation and a little bit of Perl magic to pick which character(s) to separate the words with.

- **Context:** There's a term you've heard used just recently but which hasn't been explained: context. Context refers to how an expression or variable is evaluated in Perl. The two main contexts are:
 - Scalar context
 - List context

Scalar variables are always evaluated in scalar context, however arrays and hashes can be evaluated in both scalar contexts (when we treat them as scalars) and list contexts (when we treat them as arrays and hashes). Scalar variables are always evaluated in scalar context, however arrays and hashes can be evaluated in both scalar contexts (when we treat them as scalars) and list contexts (when we treat them as arrays and hashes).

Many things in Perl are very specific about which context they require and will force lists into scalar context when required. For example the `+` (plus or addition) operator expects that its two arguments will be scalars. Hence:

```

0 my @a = (1,2,3);
1 my @b = (4,5,6,7);
2 print @a + @b;

```

will print 7. There's also a third type of context, the null context, where the result of an operation is just thrown away. This usually isn't discussed, because by its very definition we don't care about what result is returned

- **Lists:** A list is an unnamed array.

```

0 print ("Hello", ",", " ", "World", "\n");

```

If you come across something that wants a LIST, you can either give it the elements of list as in the first example above, or you can pass it an array by name. If you come across something that wants an ARRAY, you have to actually give it the name of an array. Examples of functions which insist on wanting an ARRAY are `push()` and `pop()`, which can be used for adding and removing elements from the end of an array

- **Anonymous array reference:**

```
0 my $ref = [1,2,3];
```

The `[ ... ]` syntax is used to create and assign an anonymous array reference. Square brackets `[ ]` create a memory reference to an array that has no name. This reference is a single scalar value, meaning it is stored in a standard scalar variable (`$`).

- **Multiple dimensional arrays:**

```
0 my $matrix = [  
1     [1,2,3],  
2     [4,5,6],  
3     [7,8,9]  
4 ];
```

Passing a reference to a subroutine avoids flattening your data structure and is much faster for large lists because Perl only copies the single memory address rather than the entire list.

- **Dereferencing:**

```
0 print $array_ref->[0]; # Outputs: apple  
1  
2 # Loop through all elements  
3 for my $item (@$array_ref) {  
4     print "$item\n";  
5 }
```

The syntax

```
0 @{$array_ref}
```

is used to dereference an array reference.

Hashes

- **Hashes:** A hash is a two-dimensional array which contains keys and values, they're sometimes called "associative arrays", or "lookup tables". Instead of looking up items in a hash by an array index, you can look up values by their keys.

Hashes are initialised in exactly the same way as arrays, with a comma separated list of values:

```
0 my %monthdays = ("January", 31, "February", 28, "March", 31,  
  ↪ ...);
```

Or,

```
0 my %monthdays = (  
1   "January" => 31,  
2   "February" => 28,  
3   "March" => 31,  
4   ...  
5 );
```

The => operator is syntactically the same as the comma, but is used to distinguish hashes more easily from normal arrays. It's pronounced "fat comma", or less often "fat arrow". It does have one difference, you don't need to put quotes around a bare word immediately before the => operator as these are always treated as strings:

- **Reading hash values:**

```
0 print $monthdays{"January"}; # prints 31
```

Bare words inside the braces of the hash-lookup will be interpreted in Perl as strings, so usually you can drop the quotes:

- **Creating hash elements:** You can also create elements in a hash on the fly:

```
0 my %monthdays = ();  
1 $monthdays{"January"} = 31;  
2 $monthdays{February} = 28;  
3 ...
```

- **Changing hash values:** To change a value in a hash, just assign the new value to your key:

```
0 $pizza_prices{small} = '$6.00'; # Small pizza prices have  
  ↪ gone up
```

- **Deleting hash values:** Use the delete function

```
0 delete($pizza_prices{medium}); # No medium pizzas anymore.
```

- **Finding out the size of a hash:** Strictly speaking there is no equivalent to using an array in a scalar context to get the size of a hash. If you take a hash in a scalar context you get back the number of buckets used in the hash, or zero if it is empty. This is only really useful to determine whether or not there are any items in the hash, not how many.

If you want to know the size of a hash, the number of key-value pairs, you can use the `keys` function in a scalar context. The `keys` function returns a list of all the keys in the hash.

```
0 my $size = keys %monthdays;  
1 print $size;
```


Special variables

- **Intro:** Perl has many special variables. These are used to set or retrieve certain values which affect the way your program runs. For instance, you can set a special variable to turn interpreter warnings on and off (`$^W`), or read a special variable to find out the command line arguments passed to your script (`@ARGV`).
- `$_`: Represents the current thing that your Perl script's working with — often a line of text or an element of a list or hash. It can be set explicitly, or it can be set implicitly by certain looping constructs (which we'll look at later).

The special variable `$_` is often the default argument for functions in Perl. For instance, the `print()` function defaults to printing `$_`.

```
0  $_ = "Hello, world!\n";
1  print;
```

- `@ARGV`: Command line args end up in this special array.

```
0  # ./main.pl arg1 arg2
1  print "@{ARGV}" arg1 arg2
2  print my $argc = @ARGV; 2
```

- `%ENV`: This hash contains the names and values of environment variables.
- `!`: `!` contains the operating-system error corresponding roughly to `errno` in C.

```
0  open(my $fh, "<", "foo.txt")
1      or die "open failed: !";
```

- **Input record separator `$/`:** Normally,

```
0  $/ = "\n";
```

- **Proc ID `$$`:** When used by itself without a variable name, `$$` is a predefined global variable that returns the Process ID (PID) of the currently running Perl script.
- **List of general special variables:**
 - `$_` Default input / working variable
 - `@_` Arguments passed to the current subroutine
 - `$"` Separator used when interpolating arrays into strings
 - `$$` Current process ID
 - `$0` Name of the running Perl program
 - `$(` Real group ID
 - `$)` Effective group ID
 - `$<` Real user ID

- \$> Effective user ID
- \$; Multidimensional array subscript separator; largely obsolete
- \$a, \$b Special comparison variables used by sort
- %ENV Process environment variables
- \$] Perl version as a numeric value
- \$^V Perl version as a version object
- \$^O Operating system Perl was built for
- \$^X Perl executable used to start the program
- \$^T Program start time
- \$^F Maximum system file descriptor
- \$^I In-place editing extension (-i)
- \$^M Emergency memory pool
- @F Autosplit fields when using -a
- @INC Module/library search paths
- %INC Files/modules already loaded
- \$INC Current @INC hook index in newer Perl versions
- @ISA Parent classes of a package
- %SIG Signal handlers
- %{\^HOOK} Hooks for certain Perl operations
- \${^MAX_NESTED_EVAL_BEGIN_BLOCKS} Limit for nested eval/BEGIN processing

- **Regex variables**

- \$1, \$2, \$3, ... Captured groups
- @{\^CAPTURE} All capture-buffer contents
- \$& Entire matched string
- \$' Text before the match
- \$' Text after the match
- \$+ Highest-numbered capture that matched
- \$^N Most recently closed capture
- @- Starting offsets of match/captures
- @+ Ending offsets of match/captures
- %- Named captures, including multiple captures with the same name
- %+ Named captures
- \${\^MATCH} Entire /p match
- \${\^PREMATCH} Text before a /p match
- \${\^POSTMATCH} Text after a /p match
- \${\^LAST_SUCCESSFUL_PATTERN} Last successful regex pattern
- \$^R Result of regex (?{ ... }) evaluation
- \${\^RE_COMPILE_RECURSION_LIMIT} Regex compile nesting limit
- \${\^RE_DEBUG_FLAGS} Regex debugging flags
- \${\^RE_TRIE_MAXBUF} Regex trie optimization memory control

- **Command-line and filehandle variables:**
 - @ARGV Command-line arguments
 - \$ARGV Current filename being processed by <>
 - ARGV Special filehandle used by <>
 - ARGVOUT Output filehandle used by in-place editing
 - \$. Current input line number
 - \$/ Input record separator
 - \$ Output record separator
 - \$, Output field separator
 - \${^LAST_FH} Last filehandle read
- **Error and process-result variables:**
 - \$! OS/C-library error (errno)
 - %! Symbolic access to errno values
 - \$@ Exception from the most recent eval
 - \$? Exit/wait status of the last external process
 - \$^E Extended OS-specific error information
 - \${^CHILD_ERROR_NATIVE} Native child-process status
 - \$^S Interpreter/evaluation state
 - \$^W Global warning flag
 - \${^WARNING_BITS} Lexically enabled warning categories

More on strings

- **Foldcase:** case folding is performed using the built-in `fc` function (introduced in Perl 5.16). Case folding converts strings into a canonical format meant for case-insensitive comparisons, handling complex Unicode character mappings.

```
0 use feature 'fc'; # or use v5.16;
```

```
0 $a = 'H';  
1 $b = "h";  
2  
3 print fc($a) cmp fc($b); # 0
```

- **String escape sequences:**
 - `\u` (Uppercase next): Capitalizes only the very next character.
 - `\l` (Lowercase next): Lowers only the very next character.
 - `\U` (Uppercase all): Capitalizes all characters that follow it.
 - `\L` (Lowercase all): Lowers all characters that follow it.
 - `\F` (Foldcase all): Case-folds all characters that follow it (ideal for Unicode case-insensitive matching).
 - `\E` (End modification): Stops the effect of `\U`, `\L`, or `\F`.

Operators and functions

- **Operators**

- **Arithmetic:** +, -, *, /, %, ** (exp), +=, -=, *=, /=, %, **=

Note: In fact, the shortcut operators += and its siblings can be used for just about any operator.

We also have post/pre increment/decrement.

- **Logical:** &&, and, ||, or, !, not, xor, //

Note: The versions and, or, not, xor have extremely low precedence compared to &&, ||, and !, lower even than the assignment operator =.

The // operator is defined-or, which returns the left side if it is defined, even if it is false (like 0 or "").

- **Numeric conditional:** ==, !=, <, >, <=, >=, <=>
- **String conditional:** eq, ne, lt, gt, le, ge, cmp

Note: <=> and cmp perform three-way sorting and evaluation. They return -1, 0, or 1 for less, equal, or greater.

- **String:** . (concat), x (repeat)

```
0  "hello" . "world";
1  "hello" x 3;
```

- **Ternary:** Uses C syntax

```
0  condition ? if_true : if_false
```

- **Function calling:** Perl does not insist that functions enclose their arguments within parentheses. Both print "Hello"; and print("Hello"); are correct.

- **Argument types:**

- **SCALAR** – Any scalar variable, for example: 42, "foo", or \$a.
- **EXPR** – An expression (possibly built out of terms and operators) which evaluates to a scalar.
- **LIST** – Any named or unnamed list (remember that a named list is an array).
- **ARRAY** – A named list; usually results in the array being modified.
- **HASH** – Any named or unnamed hash.
- **PATTERN** – A pattern to match on — we'll talk more about these later on, in Regular Expressions.
- **FILEHANDLE** – A filehandle indicating a file that you've opened or one of the pseudo-files that is automatically opened, such as STDIN, STDOUT, and STDERR.

- **Data conversion functions:**

- **hex:** Interprets a string as a hexadecimal number and converts it to decimal.
- **oct:** Interprets a string as an octal number and converts it to decimal. If the string starts with 0x, it automatically parses it as hex, or with 0b as binary.
- **int:** Strips the decimal portion of a floating-point number, truncating it to an integer.
- **chr:** Converts an ASCII/Unicode numeric value into its corresponding character.
- **ord:** The inverse of chr. Converts the first character of a string into its numeric ASCII/Unicode value.
- **join:** Flattens a list or array into a single string, using a specified separator string.

```
o my $string = join("-", ("a", "b", "c")); # Returns
  ↪ "a-b-c"
```

- **split:** The inverse of join. Breaks a string into a list of substrings using a regular expression delimiter.

```
o my @array = split(/-/ , "a-b-c"); # Returns ('a', 'b',
  ↪ 'c')
```

9.1 Quote operators and siblings

- **q// (Single Quote):** Behaves exactly like standard single quotes ('...'). It does not interpolate variables or escape sequences.
- **qq// (Double Quote):** Behaves exactly like standard double quotes ("..."). It interpolates variables, math, and escape sequences (like \n or \U).
- **qw// (Quote Words):** Splits a string on whitespace and returns a list of literal words. No variable interpolation occurs

9.2 Important builtin functions

- **Numeric:**

- `abs($x)` Absolute value
- `sqrt($x)` Square root
- `exp($x)` $e^{**} \$x$
- `log($x)` Natural logarithm
- `sin($x)` Sine
- `cos($x)` Cosine
- `atan2($y, $x)` Arctangent
- `rand($n)` Random number from 0 to $< \$n$
- `srand()` Seed random number generator

- **String:**

- `length($str)` String length
- `substr($str, ...)` Extract/replace substring
- `index($str, $sub)` Find substring
- `rindex($str, $sub)` Find substring from right
- `lc($str)` Lowercase entire string
- `uc($str)` Uppercase entire string
- `fc($str)` foldcase entire string
- `lcfirst($str)` Lowercase first character
- `ucfirst($str)` Uppercase first character
- `chomp($str)` Remove input record separator, usually newline
- `chop($str)` Remove final character
- `sprintf(...)` Produce formatted string

```
0  sprintf(FORMAT, LIST);  
1  
2  my $price = 19.9962;  
3  my $money = sprintf("%.2f", $price);
```

- `split(...)` Split string into list
- `join(...)` Combine list into string

- **Arrays:**

- `push(@a, $x)` Add to end
- `pop(@a)` Remove from end
- `unshift(@a, $x)` Add to beginning
- `shift(@a)` Remove from beginning
- `splice(@a, ...)` Insert/remove array elements
- `reverse(@a)` Reverse elements
- `sort(@a)` Sort elements
- `grep ... @a` Filter elements
- `map ... @a` Transform elements
- `join($sep, @a)` Convert array to string
- `scalar(@a)` Number of elements
- `each(@a)` Iterate index/value pairs on modern Perl


```

0  my @nums = (3, 1, 2);
1
2  push @nums, 4;
3
4  my $last = pop @nums;
5
6  my @sorted = sort { $a <=> $b } @nums;
7
8  my @even = grep { $_ % 2 == 0 } @nums;
9
10 my @doubled = map { $_ * 2 } @nums;

```

- **Hashes:**

- keys(%h) Get keys
- values(%h) Get values
- each(%h) Get key/value pairs while iterating
- exists(%h{\$key}) Check whether key exists
- delete(%h{\$key}) Remove key
- scalar(keys %h) Number of keys

Constructs

- **If**

```
0  if (cond) {  
1      ...  
2  }  
3  
4  elsif (cond) {  
5      ...  
6  }  
7  
8  else {  
9      ...  
10 }
```

- **Unless:** inverse of an if

```
0  unless (cond) {  
1      ...  
2  }
```

Note that the brackets are required, and there is no `elsunless`.

- **Shorthand:** If if or unless require a single statement, we can put the statement before the check

```
0  print "Hello" if 1;
```

- **Existence and definitiveness:** To find out if something is defined, use Perl's defined function. The defined function is necessary to distinguish between a value that is false because it is undefined and a value that is false but defined, such as 0 (zero) or "" (the empty string).

```
0  my $skippy; # $skippy is undefined and false  
1  $skippy = "bush kangaroo"; # true and defined  
2  print "true" if $skippy; # prints true  
3  print "defined" if defined($skippy); # prints defined  
4  
5  my $undef = undef;  
6  print "defined" if defined($undef); # does not print  
7  print "undefined" if not defined($undef); # prints undefined
```

- **undef:** When you pass a variable or expression to it, undef behaves as a function that un-defines the variable and frees its memory. The expression must be an lvalue

```
0  undef $scalar;          # Resets a scalar to the undefined state
```

When used on its own without any arguments, it simply returns the undefined value.

```
0 my $name = undef;    # Explicitly assigning the undefined
  ↪ value
```

- **While:**

```
0 while (cond) {
1     ...
2 }
```

- **Until:** The opposite of a while

```
0 until (cond) {
1     ...
2 }
```

- **Shorthand while / unless:** Similar to if and unless,

```
0 print "Hello" while 1;
1 print "Hello" until 0;
```

- **For:** Identical to C

```
0 for (my $i = 0; $i < $n; ++$i)
```

- **Foreach:**

```
0 foreach(@arr) {
1     print "$_\n";
2 }
3
4 print "${_}\n" foreach(@arr);
```

Note: When in a foreach loop, the variable representing the current element is the current element, not a copy of it.

- **Foreach range syntax:**

```
0 print "${_}" foreach (1..10)
```

- **For / foreach and variable creation**

```

0  for my $i (0..10) { ... }
1  foreach my $i ((1,2,3)) {...}

```

- **Note about for and foreach:** In Perl, you can iterate through an array using the for loop (or foreach loop). Perl treats for and foreach as completely interchangeable synonyms.

```

0  my @arr = qw(1 2 3);
1  print "$_\n" for @arr;

```

- **= do ...:**

```

0  my $variable = do { ... };

```

executes a block of code and returns the value of the last statement evaluated inside that block

- **eval BLOCK:**

```

0  eval {...}

```

acts like a try-catch block.

```

0  # Example of trapping a division by zero error
1  eval {
2      my $result = 10 / 0;
3  };
4
5  if ($?) {
6      print "Error caught: $@";
7  }

```

- **eval EXPR:** This form compiles code string text dynamically

```

0  # Example of dynamically executing a string
1  my $code = '$x = 5 + 5;';
2  eval $code;
3  print $x; # Outputs 10

```

- **Labeled blocks:** You can assign a name (called a label) to any code block by placing the name followed by a colon (:) right before the opening brace. This allows you to jump out of or control specific nested blocks using loop control statements like last, next, or redo.

Always write labels in ALL CAPS to ensure they stand out and never conflict with reserved Perl keywords.

```

0  OUTER_LOOP: while ($x < 10) {
1      INNER_LOOP: while ($y < 10) {
2          if ($condition) {
3              # Exits BOTH loops completely, not just the inner
3              ↪ one
4              last OUTER_LOOP;
5          }
6          $y++;
7      }
8      $x++;
9  }

```

- redo restarts the current iteration immediately.
 - next skips to the next iteration
 - last exits the loop completely (like break in C/Java).
- **BEGIN:** A BEGIN block in Perl is a special type of code block that runs as soon as it is parsed and compiled, before the rest of the surrounding program runs.
 - **Timing:** It executes immediately during the compilation phase, even before the rest of the script or file is fully compiled.
 - **Order:** Multiple BEGIN blocks run in the order they appear in your code.
 - **Purpose:** Commonly used to load modules (use is implemented using BEGIN), define subroutines, or set up global configuration variables before other code compiles.
 - **Automatic Deletion:** Once a BEGIN block has run, it is immediately deleted from memory for the rest of the program

```

0  print "2. This runs during runtime.\n";
1
2  BEGIN {
3      print "1. This runs first, during compilation.\n";
4  }

```

```

0  1. This runs first, during compilation.
1  2. This runs during runtime.

```

- **END:** END {} block in Perl runs code automatically just before the Perl interpreter exits the program.
 - **Timing:** It runs as late as possible during program shutdown, acting like a destructor.
 - **Order:** Multiple END blocks run in the reverse order of how they appear in your code.
 - **Exceptions:** END blocks run even if the program exits using die(). They do not run if the program exits via exec or a hard operating system signal like SIGKILL.
 - **Common Uses:** They are great for cleanup tasks, such as closing file handles, removing temporary files, or disconnecting from databases.

Input / output

- **Predefined file handles:**

```
0  STDIN    # standard input
1  STDOUT   # standard output
2  STDERR   # standard error
```

- **<...> syntax:** Primarily a read operation on a filehandle

```
0  my $line = <STDIN>;
```

Read the next line from standard input.

- **Writing to file handle:**

```
0  print STDOUT "Hello\n";
```

Although STDOUT is the default.

- **readline:** Reads a line from a file handle

```
0  my $line = readline(STDIN);
```

Reads a line from

- **say and printf:** Say is essentially C puts; takes a single string argument and appends a newline before printing it.

```
0  use feature "say";
1  say "Hello, world";
2  printf "%s", "Hello, world";
```

- **Opening a file:**

```
0  open(HANDLE, MODE, PATH);
```

- < read
- > write, truncate existing file
- » append
- +< read and write (starts at beginning)
- +> read and write, truncate
- +» read and append (starts at end)

- **Closing a file:**

```
0  close(HANDLE);
```

- **Empty diamond:** The empty diamond operator reads from all files specified on the command line (in order).

```
0  while (<>) {  
1    print;  
2  }
```

- **Slurp mode:** We can temporarily make the input separator undefined, then read a file.

```
0  open(my $file, "<", "file.txt")  
1    or die "$!";  
2  
3  my $contents = do {  
4    local $/  
5    <$file>;  
6  }
```

This is called **file slurping**

Subroutines

- Creating and calling subroutines

```
0  sub fn {  
1      ...  
2  }  
3  
4  fn(); # Preferred method of calling  
5  &fn();  
6  &fn;  
7  fn;
```

The older ampersand syntax bypasses any defined prototypes and implicitly passes the caller's current `@_` argument array.

You can pass arguments to a subroutine by including them in the parentheses when you call it. The arguments end up in a special array called `@_` which is only visible inside the subroutine.

- **Shift and unshift:** The Perl shift function removes and returns the first element of an array. This shortens the array by one and shifts all subsequent elements to the left. If the target array is empty, it returns undef

The unshift function adds one or more elements directly to the beginning of an array. It is the exact opposite of the shift function, which removes an item from the front.

```
0  sub print_headers {  
1      my $title = shift "Untitled";  
2      my $author = shift "Anonymous";  
3      print "$title\n"  
4      print "by\n"  
5      print "$author\n";  
6  }
```

- **Passing arrays and hashes:** To pass in a single array or hash to a subroutine, make it the final element in your argument list.

Passing in more than one array or hash causes problems. This is because of list flattening. When Perl sees a number of items in parentheses these are combined into one big list.

```
0  # Flatten two lists into one big list and put that in an  
    ↪ array  
1  my @biglist = (@list1, @list2);  
2  
3  # Flatten (join) two hashes into one big list an put that in  
    ↪ a hash  
4  my %bighash = (%hash1, %hash2);
```



```

0  my @colours = qw/red blue white green pink/;
1  my @chosen = qw/red white green/;
2
3  print_unchosen(@chosen, @colours);
4
5  sub print_unchosen {
6      my (@chosen, @colours) = @_;
7
8      # at this point @chosen contains:
9      # (red white green red blue white green pink)
10     # and @colours contains () - the empty list.
11 }

```

We can avoid this problem by using references:

```

0  my @colours = qw/red blue white green pink/;
1  my @chosen = qw/red white green/;
2  print_unchosen(\@chosen, \@colours);
3  sub print_unchosen {
4      my ($chosen, $colours) = @_;
5      my @chosen = @$chosen;
6      my @colours = @$colours;
7      # at this point @chosen contains:
8      # (red white green)
9      # and @colours contains (red blue white green pink)
10 }

```

- **Default arguments:** Starting in Perl v5.36, subroutine signatures are stable and enabled automatically when you specify `use v5.36;` or higher. You can define default values directly inside the parameter list using `=`, `//=`, or `||=`.

```

0  use v5.36;
1
2  # 1. Standard Default (=)
3  # Triggered only if the argument is omitted entirely
4  sub greet ($name, $greeting = "Hello") {
5      say "$greeting, $name!";
6  }
7  greet("Alice");           # Hello, Alice!
8  greet("Bob", "Good morning"); # Good morning, Bob!
9  greet("Charlie", undef);  # Prints: , Charlie! (undef treated
   ↪ as a value)
10
11 # 2. Defined-or Default (//=) - Added in v5.38+
12 # Triggered if omitted OR passed as undef
13 sub set_timeout ($time //= 30) {
14     say "Timeout is $time seconds.";
15 }
16 set_timeout();           # Timeout is 30 seconds.
17 set_timeout(undef);      # Timeout is 30 seconds.
18
19 # 3. Logical-or Default (=) - Added in v5.38+
20 # Triggered if omitted, undef, or any false value (0, "")
21 sub set_status ($status = "active") {
22     say "Status: $status";
23 }
24 set_status(0);           # Status: active (0 is false)

```

Because Perl does not strictly enforce the count of elements in `@_` natively, all positional arguments are technically optional unless you explicitly check them.

```

0  sub connect_db {
1      # Unpack parameters from @_
2      my ($host, $port, $timeout) = @_;
3
4      # Assign fallback values if they aren't provided (or are
   ↪ undef)
5      $port    //= 3306;
6      $timeout //= 30;
7
8      print "Connecting to $host:$port (Timeout: $timeouts)\n";
9  }
10
11 connect_db("localhost");           # Uses defaults for port
   ↪ and timeout
12 connect_db("127.0.0.1", 8080);     # Override port, uses
   ↪ default timeout

```

- **Named arguments:** When your subroutine accepts many optional parameters, remembering the exact positional order becomes difficult. The standard Perl convention is to slurp remaining arguments into a hash, allowing users to pass named parameters

```

0  use v5.10; # For the // operator
1
2  sub create_user {
3      my ($username, %options) = @_; # First argument is
        ↪ required, rest are optional pairs
4
5      # Set fallback defaults
6      my $role   = $options{role}    // 'subscriber';
7      my $status = $options{status} // 'active';
8
9      print "Created $username as a $role ($status)\n";
10 }
11
12 # Callers can pass options in any order, or skip them
        ↪ entirely
13 create_user("jdoe");
14 create_user("msmith", role => "admin");
15 create_user("rrock", status => "pending", role =>
        ↪ "moderator");

```

- **Ampersand fn name:** If you call a subroutine using the ampersand and omit parentheses (e.g., `&my_sub`), Perl automatically passes the caller's current argument array (`@_`)

Also, if you name a custom subroutine the same name as a built-in Perl function or keyword, the ampersand forces Perl to run your custom subroutine instead of the built-in version.

- **Subroutine references:** When paired with a backslash (`&my_sub`), it does not call the function. Instead, it creates a reference to the subroutine, which allows you to pass the function around as data

When you use parentheses, it clears the argument list.

- **Function prototypes:** In Perl, function prototypes are a mechanism used to change how the Perl parser interprets a subroutine's arguments, primarily allowing user-defined functions to mimic built-in Perl functions.

Unlike prototypes in languages like C, C++, or Java, Perl prototypes do not act as traditional function signatures. Instead of validating types at runtime, they force context coercion on variables and change how arguments are parsed at compile time

Prototypes are declared by adding special characters inside parentheses right after the subroutine name:

- `$` Forces scalar context on the argument.
- `@` Consumes all remaining arguments as a list (slurpy).

An `@` prototype is slurpy, so it must appear at the end of the prototype:

- `%` Consumes all remaining arguments as a hash.

Internally, Perl still passes a flat list:

The % prototype causes the remaining arguments to be evaluated in list context. It does not guarantee that the arguments form a valid hash, so an odd number of arguments is still possible.

It can follow mandatory arguments:

```
0 sub configure ($%) {  
1     my ($name, %options) = @_;  
2 }  
3  
4 configure(  
5     "window",  
6     width => 800,  
7     height => 600,  
8 );
```

- & Requires a code block or anonymous subroutine reference.

When & is the first prototype character, the caller can pass a bare block:

```
0 sub repeat (&$) {  
1     my ($code, $count) = @_;  
2  
3     $code->() for 1 .. $count;  
4 }  
5  
6 repeat {  
7     print "Hello\n";  
8 } 3;
```

This is approximately equivalent to:

```
0 repeat(sub {  
1     print "Hello\n";  
2 }, 3);
```

- * Accepts a typeglob. The * prototype is commonly used for filehandles:

```
0 sub write_message (*$) {  
1     my ($handle, $message) = @_;  
2     print $handle "$message\n";  
3 }  
4  
5 write_message(STDOUT, "Hello");  
6 write_message(STDERR, "An error occurred");
```

It can also accept a typeglob explicitly:

```
0 write_message(*STDOUT, "Hello");
```

The `*` prototype supports Perl's bareword filehandle syntax. It does not strictly enforce that the value is a typeglob at runtime.

- `;` Separates mandatory arguments (before `;`) from optional ones (after `;`).

```
0 sub greet ($;$) {  
1     my ($name, $greeting) = @_;  
2  
3     $greeting //= "Hello";  
4  
5     print "$greeting, $name!\n";  
6 }
```

- `\_` Requires a literal variable starting with that character and passes it by reference. It means that the caller must provide an actual variable of the corresponding kind. Perl then passes a reference to it automatically.

Note: The subroutine must be declared before its calls for the prototype to take effect.

Also, you cannot write something like

```
0 sub _display(first, @) { ... }
```

because a Perl prototype describes the context and syntactic form of each argument. It does not declare or name parameters.

The characters inside the parenthesis must be entirely prototype characters. To require one scalar argument followed by any number of additional arguments, write:

```
0 sub _display($@) { ... }
```

Modern Perl also supports actual named parameter declarations through signatures:

```
0 use v5.36;  
1 sub _display($first, @remaining) { ... }
```

Here, `($first, @remaining)` is a signature, not a prototype. It creates the lexical variables `$first` and `@remaining` automatically.

Use signatures when your goal is to name and unpack parameters. Use prototypes only when you intentionally want to alter how Perl parses or evaluates calls to the subroutine.

- **Bypassing prototype checking (ampersand prefix):** If a subroutine has a defined prototype (which restricts or formats the arguments it accepts), prefixing the call with an ampersand will completely ignore the prototype rules.

Regular expressions

- **Matching operator:** The most basic regular expression operator is the matching operator, `m/PATTERN/`.
 - Works on `$_` by default.
 - In scalar context, returns true (1) if the match succeeds, or false (the empty string) if the match fails.
 - In list context, returns a list of any parts of the pattern which are enclosed in parentheses. If there are no parentheses, the entire pattern is treated as if it were parenthesised.
 - The `m` is optional if you use slashes as the pattern delimiters.
 - If you use the `m` you can use any delimiter you like instead of the slashes. This is very handy for matching on strings which contain slashes, for instance directory names or URLs.
 - Using the `/i` modifier on the end makes it case insensitive.

```
0  while (<>) {  
1      print if m/foo/;  
2  }
```

- **s/pattern/replacement:** This is the substitution operator, and can be used to find text which matches a pattern and replace it with something else.
 - Works on `$_` by default.
 - In scalar context, returns the number of matches found and replaced.
 - In list context, behaves the same as in scalar context and returns the number of matches found and replaced (a cause of more than one mistake...).
 - You can use any delimiter you want, the same as the `m//` operator.
 - Using `/g` on the end of it matches globally, otherwise matches (and replaces) only the first instance of the pattern.
 - Using the `/i` modifier makes it case insensitive.
- **Binding operators:** If we want to use `m//` or `s///` to operate on something other than `$_` we need to use binding operators to bind the match to another string.
 - `=~` True if the pattern matches
 - `!~` True if the pattern doesn't match

The most common use is to test if a string matches a regular expression. It returns a true value if it matches, and false if it doesn't.

```
0  my $text = "The quick brown fox jumps over the lazy dog";  
1  
2  # Evaluates to true because "fox" is in the string  
3  if ($text =~ /fox/) {  
4      print "Found a fox!\n";  
5  }
```

You can use `=~` with the `s///` operator to find a pattern and replace it. This modifies the original variable on the left

```
0 my $greeting = "Hello World";
1
2 # Replaces "World" with "Perl"
3 $greeting =~ s/World/Perl/;
4
5 print $greeting; # Outputs: Hello Perl
```

When you use matching parentheses `()` in your regular expression, `=~` can be used to extract those specific pieces into variables.

```
0 my $date = "2026-08-20";
1
2 # Captures the digits into $1, $2, and $3
3 if ($date =~ /(\d{4})-(\d{2})-(\d{2})/) {
4     print "Year: $1, Month: $2, Day: $3\n";
5 }
```

does not match a pattern, use the `!~` operator instead.

```
0 my $email = "user-at-domain.com";
1
2 if ($email !~ /@/) {
3     print "Invalid email: Missing '@' symbol.\n";
4 }
```

In Perl, `$1`, `$2`, `$3` (and so on) are special variables that hold captured text from the most recent successful regular expression match.

- **Modifiers:**

- `/i` Make match/substitute match case insensitive
- `/g` Make substitute global (all occurrences are changed)
- `/m` – Multi-line mode. Causes `^` and `$` to match the start and end of any line within the string rather than just the absolute string boundaries.
- `/s` – Single-line mode. Treats the string as a single line, allowing the dot `(.)` metacharacter to match newline `(\n)` characters.
- `/x` – Extended legibility. Tells Perl to ignore unescaped whitespace and allow comments `(#)` inside the regex pattern for clearer layout formatting
- `/e` – Evaluate replacement. Treats the replacement side as a block of Perl code to execute, utilizing its return value as the final replacement text
- `/ee` – Double evaluate. Evaluates the string on the replacement side as a Perl string, and then runs `eval` a second time on the result.
- `/r` – Non-destructive substitution. Instead of modifying the original variable, it performs the operation and returns the mutated string copy.

- /a – ASCII-only restrictions. Forces character classes like \d, \s, and \w to strictly match ASCII characters only.
- /u – Unicode rules. Forces character classes to abide by full Unicode rules (the default setting for most modern Perl scripts).
- /l – Locale rules. Instructs Perl to use the underlying system’s runtime locale setting.
- /d – Default rules. Uses old, backwards-compatible platform defaults.

You can also activate or deactivate modifiers dynamically inside a regex pattern using the (?imsx-imsx) synt

- (?i)apple matches "Apple" or "APPLE" inline.
- (?-i)apple turns off case-insensitivity from that point forward.

- **Meta characters:**

- ^ Start of string
- \$ End of string
- . Any single character except \n
- \n Newline
- \t Matches a tab
- \s Any whitespace character, such as space, tab, or newline
- \S Any non-whitespace character
- \d Any digit (0 to 9)
- \D Any non-digit
- \w Any "word" character - alphanumeric plus underscore (_)
- \W Any non-word character
- \b A word break - the zero-length point between a word character (as defined above) and a non-word character.
- \B A non-word break - anything other than a word break

- **Quantifiers**

- ? 0 or 1
- * 0 or more
- + 1 or more
- {n} match exactly n times
- {n,} match n or more times
- {n,m} match between n and m times

- **Character classes:** Created the usual way with brackets

- ^ negates the class
- - specifies a range of characters. To mach a literal -, it must be either the first or the last character in the class.
- \$, ., (), {}, *, + and other meta characters taken literally

- **Match variable (\$&):** This special variable holds whatever the last regex matched.

- **Alternation:** Alternation is done the usual way


```
0 /^(rabbit|chicken|dog)$/
```

- **Atoms:** In regular expressions, an atom is the most basic, single unit of a pattern match. It either matches a specific set of characters or positions, and quantifiers (like `*` or `+`

We use it to mean "something that is a chunk of regular expression in its own right". Atoms can be arbitrarily created by simply wrapping things in round brackets.

- **Literal characters:** A single ordinary letter or number (like `a` or `7`).
- **Character classes:** Brackets matching any single character inside them (like `[abc]` or `[\^0-9]`).
- **The wildcard dot:** The `.` symbol matching any character.
- **Escaped characters:** Special shorthands (like `\d`, `\w`, or `\s`).
- **Grouped expressions:** Parentheses surrounding a sub-pattern (like `(abc)`) treated as one combined unit.

References and complex data structures

- **Intro:** There are three main uses for Perl references
 1. **Creating complex data structures:** Perl references can be used to create complex data structures, for instance hashes of arrays, arrays of hashes, hashes of hashes, and more
 2. **Passing arrays and hashes to subroutines and functions:** Since all arguments to subroutines are flattened to a list of scalars, it is not possible to use two arrays as arguments and have them retain their individual identities without using references.
 3. **Object-oriented Perl:** References are used extensively in object oriented Perl. In fact, Perl objects are references to data structures.
- **Creating and dereferencing references:** To create a reference to a scalar, array or hash, we prefix its name with a backslash:

```
0 my $scalar = "This is a scalar";
1 my @array = qw(a b c);
2 my %hash = (
3     'sky' => 'blue',
4     'apple' => 'red',
5     'grass' => 'green',
6 );
7
8 my $scalar_ref = \$scalar;
9 my $array_ref = \@array;
10 my $hash_ref = \%hash;
```

Note that all references are scalars, because they contain a single item of information: the memory address of the actual data

- **ref():** You can find out whether a scalar is a reference or not by using the `ref()` function, which returns a string indicating the type of reference, or `undef` if the scalar is not a reference.
- **Dereferencing:** Dereferencing (getting at the actual data that a reference points to) is achieved by prepending the appropriate sigil to the name of the reference. For instance, if we have a hash reference `$hash_reference` we can dereference it by adding a percentage sign: `%%hash_reference`.
- **Arrow notation:** Uses the C arrow concept for a dereferenced access.

```
0 print $$array_ref[0];
1 print $array_ref->[0];
2 print $hash_ref->{"sky"};
```

- **Anonymous data structures:** We can use anonymous data structures to create complex data structures without having to declare many temporary variables. Anonymous arrays are created by using square brackets instead of round ones. Anonymous hashes use curly braces instead of round ones

```

0 my $array_ref = ['a', 'b', 'c', 'd'];
1 my $array_ref = [qw(a b c d)];
2
3 my %transport = (
4     'cars' => [qw(toyota ford holden porsche)],
5     'planes' => [qw(boeing harrier)],
6     'boats' => [qw(clipper skiff dinghy)],
7 );
8
9 my $hash_ref = {
10     a => 1,
11     b => 2,
12     c => 3
13 };

```

Data is pulled out of an anonymous data structure using the arrow notation:

- **Complex data structures:** References are most often used to create complex data structures. Since references are scalars, they can be used as values in both hashes and arrays. This makes it possible to create both deep and complex multi-dimensional data structures.

The use of references in data structures allows you to create arrays of arrays, arrays of hashes, hashes of arrays and hashes of hashes

- **Disambiguation and curly braces:** Consider

```

0 my @res = @$array[0];

```

Does this mean:

- Find @array.
- Look up index 0: \$array[0]
- Turn that (\$array[0]) into an array: @\$array[0]

Or:

- Find the array reference \$array
- Treat that as an array: @\$array
- Take an array slice with index 0: @\$array[0]

Perl does the latter, however if that is what we wanted then we should have written \$\$array[0], as that explicitly returns a single (scalar) result.

We can force Perl to evaluate our expression as the first interpretation above by using curly braces. This allows us to clearly write:

```

0 my @result = @{$array[0]};

```

We can use `${...}`, `@{...}` or `%{...}` syntax to evaluate any expression and dereference the result.

- **Data::Dumper:** Often data structures are very complex, and negotiating and printing these structures can be a tiresome exercise. It's also an unnecessary one, as all the hard work has already been done for you. To save you from having to write specialised printing code in every program for debugging purposes, there's a special library you may find useful called `Data::Dumper`.

`Data::Dumper` provides a function which takes Perl data structures and turns them into human readable strings representing the data within them. It can be used just like this:

```
0 use Data::Dumper;
```

```
0 my %HoH = (  
1     Jacinta => {  
2         age => 26,  
3         favourite_colour => "blue",  
4         sport => "swimming",  
5         language => "Perl",  
6     },  
7  
8     Paul => {  
9         age => 27,  
10        favourite_colour => "green",  
11        sport => "cycling",  
12        language => "Perl",  
13    },  
14 );  
15 print Dumper \%HoH;
```

This will print out something similar to:

```
0 $VAR1 = {  
1     'Paul' => {  
2         'language' => 'Perl',  
3         'favourite_colour' => 'green',  
4         'sport' => 'cycling',  
5         'age' => 27  
6     },  
7  
8     'Jacinta' => {  
9         'language' => 'Perl',  
10        'favourite_colour' => 'blue',  
11        'sport' => 'swimming',  
12        'age' => 26  
13    }  
14 };
```

Not only is this easy to read, but it's also perfectly valid Perl code. This means you can use `Data::Dumper` to easily give you a structure that you can paste into another program, or which can be 'serialised' to a file and re-created at a later date. `Data::Dumper` has a lot more uses beyond simple debugging.

`Dumper` expects to be given one or more references to data structures to dump. If `Dumper` is provided with a hash or array then every element of the array, or every key and value of the hash, will be considered a separate data structure, and dump separately. The results are not particularly useful:

```
0  # result of: print Dumper %HoH;
1  $VAR1 = 'Paul';
2  $VAR2 = {
3      'language' => 'Perl',
4      'favourite_colour' => 'green',
5      'sport' => 'cycling',
6      'age' => 27
7  };
8  $VAR3 = 'Jacinta';
9  $VAR4 = {
10     'language' => 'Perl',
11     'favourite_colour' => 'blue',
12     'sport' => 'swimming',
13     'age' => 26
14  };
```

External Files and Packages

- **Require:** Perl implements a number of mechanisms for loading code from external files. The most simplest of these is by using the require function:

```
0  require 'file.pl';
```

Perl is smart enough to make sure that the same file will not be included twice if it's required through the same specified name.

```
0  # The file is only included once in the following case:
1  require 'file.pl';
2  require 'file.pl';
```

Required files must end with a true value. This is usually achieved by having the final statement of the file being:

```
0  1;
```

Conflicts can occur if our included file declares subroutines with the same name as those that appear in our main program. In most circumstances the subroutine from the included file takes precedence, and a warning is given.

The use of require has been largely deprecated by the introduction of modules and the use keyword. If you're writing a code library from scratch we recommend that you create it as a module. However, require is often found in legacy code and is a useful thing to understand.

Any code in the file (except for subroutines) will be executed immediately when the file is required. The require occurs at run-time, this means that Perl will not throw an error due to a missing file until that statement is reached, and any subroutines inside the file will not be accessible until after the require.

Variables declared with my are not shared between files, they are only visible inside the block or file where the declaration occurs. To share packages between files we use package variables which are covered later in this chapter.

The use of modules (which we will learn about later) allows for external files to be loaded at compile-time, rather than run-time.

- **Use strict and warnings:** Perl pragmas, such as strict and warnings are lexically scoped. Just like variables declared with my, they last until the end of the enclosing block, file or eval.

This means that you can turn strict and warnings on in one file without it influencing other parts of your program. Thus, if you're dealing with legacy code, then your new libraries, modules and classes can be strict and warnings compliant even though the older code is not.

- **Require example:**

```

0  # Greetings.pl
1  use strict;
2  use warnings;
3  my %greeting_in = (
4      en => "Hello",
5      'en-au' => "G'day",
6      fr => "Bonjour",
7      jp => "Konnichiwa",
8      zh => "Nihao",
9  );
10
11  sub hello {
12      my $language = shift "en";
13
14      my $greeting = $greeting_in{$language}
15          or die "Don't know how to greet in $language";
16
17      return $greeting;
18  }
19
20  1;

```

```

0  # program.pl
1  use strict;
2  require "Greetings.pl";
3
4  print "English: ", hello("en"), "\n"; # Prints "Hello"
5  print "Australian: ", hello("en-au"), "\n"; # Prints "G'day"

```

- **Packages:** When working with a large project, the chances of naming conflicts increases. Two entirely different files may have two different subroutines with the same name; however it is only the last one loaded that will be used by Perl. Files from different projects may be re-used in new developments, and these may have considerable name clashes. Multiple files can also make it difficult to determine where subroutines are originally declared, which can make debugging difficult.

Perl's packages are designed to overcome these problems. Rather than just putting code into separate files, code can be placed into independent packages, each with its own namespace. By ensuring that package names remain unique, we also ensure that all subroutines and variables can remain unique and easily identifiable.

A single file can contain multiple packages, but convention dictates that each file contains a package of the same name. This makes it easy to quickly locate the code in any given package.

Writing a package in Perl is easy. We simply use the package keyword to change our current package. Any code executed from that point until the end of the current file or block is done so in the context of the new package.

```

0  package Greetings;
1  use strict;
2  use warnings;
3
4  my %greeting_in = (
5      en => "Hello",
6      'en-au' => "G'day",
7      fr => "Bonjour",
8      jp => "Konnichiwa",
9      zh => "Nihao",
10 );
11
12 sub hello {
13     my $language = shift "en";
14     my $greeting = $greeting_in{$language}
15         or die "Don't know how to greet in $language";
16
17     return $greeting;
18 }
19
20 1;

```

The package that you're in when the Perl interpreter starts (before you specify any package) is called `main`. Package declarations use the same rules as `my`, that is, it lasts until the end of the enclosing block, file, or `eval`.

Perl convention states that package names (or each part of a package name, if it contains many parts) starts with a capital letter. Packages starting with lower-case are reserved for pragmas (such as `strict`).

- **Scoping operator:** Perl provides a scoping operator in the form of a pair of adjacent colons. The scoping operator allows us to refer to information inside other packages, and is usually pronounced "double-colon".

```

0  require "Greetings.pl";
1
2  print Greetings::hello("en"), "\n";
3  print Greetings::hello("jp"), "\n";
4
5  # This calls the hello() subroutine in our main package
6  # (below), printing "Greetings Earthling".
7  print hello(), "\n";
8
9  sub hello {
10     return "Greetings Earthling";
11 }

```

- **Changing variables in other packages:** Occasionally we may wish to change the value of a variable in another package. It should be very rare that we should need to do this, and it's not recommended you do so unless this is a documented feature of your package. However, in the case where we do need to do this, we use the scoping operator again.


```

0  use Carp;
1
2  # Turning on $Carp::Verbose makes carp() and croak() provide
3  # stack traces, making them identical to cluck() and
   ↪ confess().
4  # This is documented in 'perldoc Carp'.
5
6  $Carp::Verbose = 1;

```

`carp()` and `croak()` are functions from the `Carp` module that report errors from the calling code's perspective instead of inside your module

- **`carp()`**: Works like `warn()`, sending a non-fatal warning, but attributes the line number to the code that called your module.
- **`croak()`**: Works like `die()`, throwing a fatal error, but attributes the line number to the code that called your module rather than the exact line inside the function.

`cluck()` and `confess()` are error-handling functions provided by the `Carp` module that include a full stack backtrace to show exactly where and how an error occurred

- `cluck()` acts like an enhanced `warn()`. It prints the error message and the full stack trace, but allows the program to keep running.
- `confess()` acts like an enhanced `die()`. It prints the error message and the full stack trace, then terminates the program immediately.

```

0  use Carp qw(cluck confess); # Explicitly importing cluck

```

- **Main package**: There's a shorthand for accessing variables and subroutines in the main package, which is to use double-colon without a package name. This means that `::$foo` is the same as `$main::foo`
- **Package variables and our**: It is not possible to access lexically scoped variables (those created with `my`) outside of their enclosing block. This means that we need another way to create variables to make them globally accessible. These global variables are called package variables, and as their name suggests they live inside their current package. The preferred way to create package variables, under Perl 5.6.0 and above, is to declare them with the `our` statement. Of course, there are alternatives you can use with older version of Perl, which we also show here:

```

0  package Carp;
1  our $VERSION = '1.01'; # Preferred for Perl 5.6.0 and above
2
3  # use vars qw/$VERSION/; # Preferred for older versions
4  # $VERSION = '1.01';
5
6  # $Carp::VERSION = '1.01'; # Acceptable but requires that we
   ↪ then
7  # always use this full name under strict

```

In all of the cases above, both our package and external code can access the variable using `$Carp::VERSION`.

Modules

- **Module:** A module is a separate file containing Perl source code, which is loaded and executed at compile time. This means that when you write:

```
0  use CGI;
```

Perl looks for a file called CGI.pm (.pm for Perl Module), and upon finding it, loads it in and executes the code inside it, before looking at the rest of your program.

Sometimes you might choose to pass extra information to the module when you load it. Often this is to request the module create new subroutines in your namespace.

```
0  use CGI qw(:standard);
1  use File::Copy qw(copy);
```

Note the use of qw(), this is a list of words (in our case, just a single word). It's possible to pass many options to a module when you load it. In the case above, we're asking the CGI module for the :standard bundle of functions, and the File::Copy module for just the copy subroutine.

Each module has a different set of options (if any) that it will accept. You need to check the documentation of the module you're dealing with to which (if any) are applicable to your needs.

- **Require vs use:** The fundamental difference is that use happens at compile time, while require happens at runtime

The use statement is actually a built-in shortcut. Writing use MyModule; is exactly equivalent to executing

```
0  BEGIN {
1      require MyModule;
2      MyModule->import();
3  }
```

Feature	use	require
Execution Time	Compile time (processed before the rest of the script runs)	Runtime (processed only when the execution line is hit)
Implicit Import	Yes. Automatically calls the module's import method.	No. Does not call import automatically.
File Types	Only loads Perl Modules (.pm files).	Loads both libraries (.pl) and modules (.pm).
Syntax	Expects a bareword name without an extension (e.g., use File::Copy;).	Accepts a bareword or a string expression with a file extension.
Conditionals	Cannot be placed inside if blocks (it will still load at compile time).	Can be placed inside if statements for conditional loading.

- **No import method:** If you do not write an import method and do not inherit one (like from `Exporter`), your module will still load perfectly fine, but it will not export anything automatically into the caller's namespace.

Perl does not require an import method. If you use `use MyModule;` and `MyModule` has no import subroutine, Perl simply skips the step without throwing an error or warning.

Because no functions or variables are copied into the main script's namespace, you cannot call them by their short names. You must use their fully qualified package paths.

```
0  # --- MyModule.pm ---
1  package MyModule;
2  sub say_hello {
3      print "Hello from the module!\n";
4  }
5  1;
6
7  # --- main.pl ---
8  use MyModule;
9
10 # say_hello();          # This will crash! Perl doesn't know
    ↳ what say_hello is.
11 MyModule::say_hello(); # This works perfectly.
```

- **Import method:** Because writing a manual import method to manipulate symbol tables is complex, almost all Perl developers inherit the behavior from the core `Exporter` module.

```

0  package MyMath;
1  use strict;
2  use warnings;
3
4  # Import the standard 'import' method from the Exporter
   ↳ module
5  use Exporter qw(import);
6
7  # Functions in @EXPORT are automatically imported by default
8  our @EXPORT = qw(add);
9
10 # Functions in @EXPORT_OK are only imported if explicitly
   ↳ requested
11 our @EXPORT_OK = qw(multiply);
12
13 sub add {
14     my ($a, $b) = @_;
15     return $a + $b;
16 }
17
18 sub multiply {
19     my ($a, $b) = @_;
20     return $a * $b;
21 }
22
23 1; # Modules must return a true value

```

- **%EXPORT_TAGS**: Group related exportable symbols into convenient shortcut tags. This allows developers importing your module to request an entire category of functions or variables at once by using a colon prefix (:tag_name)

Every symbol listed inside

You cannot nest tag names inside @EXPORT or @EXPORT_OK

Define the groups using an array reference for each tag name.

```

0  our %EXPORT_TAGS = (
1      utils => [ qw( validate_email ) ],
2      user  => [ qw( create_user delete_user ) ],
3      log   => [ qw( log_info log_error ) ],
4      all   => [ @EXPORT, @EXPORT_OK ], # Quick shortcut
   ↳ combining everything
5  );

```

When loading the module, prepend a colon (:) to the tag name to pull in that entire group

```

0  # Import only the user management group (:user)
1  use MyModule qw(:user);
2
3  create_user(); # Works!
4  delete_user(); # Works!
5
6  # log_info(); # Fails! (Not imported)

```

- **Advanced import techniques:** You can combine tags, default exports, and exclusions using specific prefixes
 - **:DEFAULT:** Refers implicitly to everything inside your module's @EXPORT array.
 - **!:** Negates/excludes a specific symbol from being imported.

```

0  # Imports default functions, the logging tag, but EXCLUDES
   ↪ log_error
1  use MyModule qw(:DEFAULT :log !log_error);

```

- **Where does Perl look for modules?:** Perl searches through a list of directories that are determined when the Perl interpreter is compiled. You can see this list (and all the other options Perl was compiled with), by using perl -V

The list of directories which Perl searches for modules is stored in the special variable @INC. It's possible to change @INC so that Perl will search in other directories as well. This is important if you have installed your own private copy of some modules.

Of course, being Perl, there's more than one way to change @INC. Here are some of the ways to add to the list of directories inside @INC:

- Call Perl with the -I command-line switch with the location of the extra directory to search. For example:

```

0  perl -I/path/to/libs

```

This can be done either in the shebang line, or on the command-line.

- Use the lib pragma in your script to inform Perl of extra directories. For example:

```

0  use lib "/path/to/libs";

```

- Setting the PERL5LIB environment variable with a colon-separated list of directories to search. Note that if your script is running with taint checks this environment variable is ignored.

Since use statements occur before regular Perl code is executed, modifying @INC directly usually does not have the desired effect.

- **Finding installed modules:** Perl comes with many modules in its standard distribution. You can get a list of all of them by doing a perldoc perlmodlib.

Besides from the modules in the standard distribution, you can also see any other modules that were installed on your system by using `perldoc perllocal`. Generally this file only lists other modules that were installed by hand, or using one of the CPAN installers (more on this later). Modules installed through your operating system's packaging system may not appear in `perldoc perllocal`.

To find a complete list of modules available on your system, regardless of how they were installed, read the documentation provided by `perldoc -q` installed.

You can get more information on any module that you have installed by using `perldoc module_name`. For example, `perldoc English` will give you information about the `English` module. You can also use `perldoc -l module_name` to locate a particular module, and `perldoc -m module_name` to view the source of a module.

- **Using CPAN (Comprehensive Perl Achieve Network) modules:** CPAN provides more than 9,000 separate and freely available modules. This makes CPAN an excellent starting point when you wish to find modules to help solve your particular problem. However, you should keep in mind that not all CPAN modules are created equal. Some are much better documented and written than others. Some (such as the `CGI` or `DBI`) modules have become de-facto standards, whereas others may not be used by anyone except the module's author.

As with any situation when you're using third party code, you should take the time to determine the suitability of any given module for the task at hand. However, in almost all circumstances it's better to use or extend a suitable module from CPAN rather than trying to re-invent the wheel.

Many of the popular CPAN modules are pre-packaged for popular operating systems. In addition, the CPAN module that comes with Perl can make the task of finding and installing modules from CPAN much easier.

Most CPAN modules come with `README` and/or `INSTALL` files which tell you how to install the modules. This may vary between operating systems. On Unix and Unix-like operating systems the process is usually:

```
0  perl Makefile.PL
1  make
2  make test
3  make install
```

For ActiveState Perl installations (which includes most Microsoft Windows machines) the use of PPM (Programmer's Package Manager) is recommended. PPM provides a command line interface for downloading and installing pre-compiled versions of most CPAN modules.

- **Writing modules:** Modules contain regular Perl code, and for most modules the vast majority of that code is in subroutines. Sometimes there are a few statements which initialise variables and other things before any of those subroutines are called, and those get executed immediately. The subroutines get compiled and tucked away for later use.

Besides from the code that's loaded and executed, two more special things happen. Firstly, if the last statement in the module did not evaluate to true, the Perl compiler throws an exception (usually halting your program before it even starts). This is so that a module could indicate that something went wrong, although in reality this feature is almost never used. Virtually any Perl module you care to look at will end with the statement `1;` to indicate successful loading.

The other thing that happens when a module is used is that its import subroutine (if one exists) gets called with any directives that were specified on the `use` line. This is useful if you want to export functions or variables to the program that's using your module for functional programming but is almost never used (and very often discouraged) for object oriented programming.

As you've no doubt guessed by now, modules and packages often go hand-in-hand. We know how to use a module, but what are the rules on writing one? The big one is this: A module is a file that contains a package of the same name.

That's it. So if you have a package called `Tree::Fruit::Citrus::Lime`, the file would be called `Tree/Fruit/Citrus/Lime.pm`, and you would use it with `use Tree::Fruit::Citrus::Lime;`

A module can contain multiple packages if you desire. So even though the module is called `Chess::Piece`, it might also contain packages for `Chess::Piece::Knight` and `Chess::Piece::Bishop`. It's usually preferable for each package to have its own module, otherwise it can be confusing to your users how they can load a particular package.

When writing modules, it's important to make sure that they are well-named, and even more importantly that they won't clash with any current or future modules, particularly those available via CPAN. If you are writing a module for internal use only, you can start its name with `Local::` which is reserved for the purpose of avoiding module name clashes.

Note: A module is a file that contains a package of the same name.

- **Use vs require:** Perl offers several different ways to include code from one file into another. `use` is built on top of `require` and has the following differences:
 - Files which are used are loaded and executed at compile-time, not run-time. This means that all subroutines, variables, and other structures will exist before your main code executes. It also means that you will immediately know about any files that Perl could not load.
 - `use` allows for the import of variables and subroutines from the used package into the current one. This can make programming easier and more concise.
 - Files called with `use` can take arguments. These arguments can be used to request special features that may be provided by some modules.

Both methods:

- Check for redundant loading, and will skip already loaded files.
- Raise an exception on failure to find, compile or execute the file.
- Translate `::` into your systems directory separator.

Where possible `use` and Perl modules are preferred over `require`.

- **extern equivalent:** The equivalent of an `extern` global variable from C in Perl is an **our** variable declared in a shared package or module.

```

0  package MyConfig;
1  use strict;
2  use warnings;
3
4  # Declare a global variable
5  our $GlobalSetting = "Active";
6
7  1;

```

```

0  use strict;
1  use warnings;
2  use MyConfig;
3
4  # Access the extern-like global variable using its full
   ↪ package path
5  print $MyConfig::GlobalSetting;
6
7  # Or bring it into your current namespace
8  our $GlobalSetting;
9  print $GlobalSetting;

```

- **Static equivalent:** A my variable is exactly how you restrict scope in Perl, but it behaves like a C static in two distinct ways depending on where you declare it.

If you declare a my variable at the top level of a file, it is invisible to other files. This behaves exactly like a file-level static variable in C.

```

0  # Inside MyModule.pm
1  use strict;
2
3  my $file_private = 42; # Visible only inside this file
4
5  sub get_value { return $file_private; }

```

If you want a variable inside a function to keep its value between calls (like a function-level static in C), use Perl's state keyword instead of my. A my variable inside a function is reinitialized every time the function runs.

- **Warnings and strict:** When your module is used by a script, whether or not it runs with warnings depends upon whether the calling script is running with warnings turned on. You can (and should) invoke the use warnings pragma to turn on warnings for your module without changing warnings for the calling script.

Your modules should always use strict.

```

0  use strict;
1  use warnings;

```


- **Exporting and importing subroutines:** Writing your own import function for each and every module would be a tiresome and error-prone process. However, Perl comes with a module called `Exporter`, which provides a highly flexible interface with optimisations for the common case.

`Exporter` works by checking inside your module for three special data structures, which describe both what you wish to export, and how you wish to export them. These structures are

- `@EXPORT` symbols to be exported into the user's name space by default
 - `@EXPORT_OK` symbols which the user can request to be exported
 - `%EXPORT_TAGS` that allows for bundles of symbols to be exported when the user requests a special export tag.
- **@ISA:** To take advantage of `Exporter`'s import function we need to let Perl know that our package has a special relationship with the `Exporter` package. We do this by telling Perl that we inherit from `Exporter`. Our package and the rest of our program does not need to be written in an object oriented style for this to work.

Now when Perl goes looking for the import function it will first look in our package. If it can't be found there, Perl will look for a special array called `@ISA`. The contents of the `@ISA` array is interpreted as a list of parent classes, and each of these will be searched for the missing method.

To specify that this package is a sub-class of the `Exporter` module we include the following lines:

```
0  use Exporter;
1  our @ISA = qw(Exporter);
```

Note: `ISA` means is-a.

- **use base:** An alternative to adding parent modules to `@ISA` yourself is to use the `base` pragma. This allows you to declare a derived class based upon the listed parent classes. Thus the two lines above becomes:

```
0  use base qw(Exporter);
```

Here's an example of just using `@EXPORT` and `@EXPORT_OK`. Our hypothetical module, `People::Manage` is used for managing interpersonal relations.

```
0  package People::Manage;
1  use base qw(Exporter);
2  use vars qw(@EXPORT @EXPORT_OK);
3
4  @EXPORT = qw(invite $name @friends %addresses); # invite is a
   ↳ subroutine
5  @EXPORT_OK = qw(&taunt $spouse @enemies %postcodes); # so is
   ↳ taunt
```

The base pragma takes care of ensuring that the Exporter module is loaded.

The base pragma is available for all versions of Perl above 5.6.0.

Idioms

- **Representation of true and false:** Perl does not have a native true or false keyword. Instead, it determines a value's truth by evaluating what it is.

Only a few specific values are considered false in Perl:

- The number 0 (or 0.0)
- The string "0"
- The empty string ""
- undef (undefined)
- An empty list ()

Everything else is true. When you use a single logical NOT (!), Perl evaluates the operand's truth and returns an absolute boolean result.

- If you pass it something false, it returns 1 (Perl's standard "true" value).
 - If you pass it something true, it must return a "false" value. By language design, Perl's logical operators return the empty string ("") to represent false.
- **Goatse operator =()**: Used to force an expression on its right-hand side into a list context while evaluating the entire assignment in a scalar context. It returns the number of items generated by the right-hand side expression

It is most frequently used to count regex matches in global matching mode (/g) without storing all the matched text in memory:

```
0 my $text = "cat dog cat bird cat";
1 my $count = () = $text =~ /cat/g;
2 print $count; # Outputs 3
```

- **Bang bang:** The bang bang operator forces any scalar value into a strict boolean 1 or empty string "" (which Perl treats as false). It is simply two consecutive logical NOT (!) operators.

The first ! evaluates the value and flips its truthiness to a strict boolean (making a true value "" and a false value 1). The second ! flips it back.

Perl doesn't have a dedicated boolean type. If a function returns an object, a string, or a number on success, using !! cleans up the data so it only returns a clean 1 (true) or "" (false).

```
0 my $user_input = "Hello";
1
2 my $pure_boolean = !!$user_input;
3 print $pure_boolean; # Outputs 1
```

Perl examples